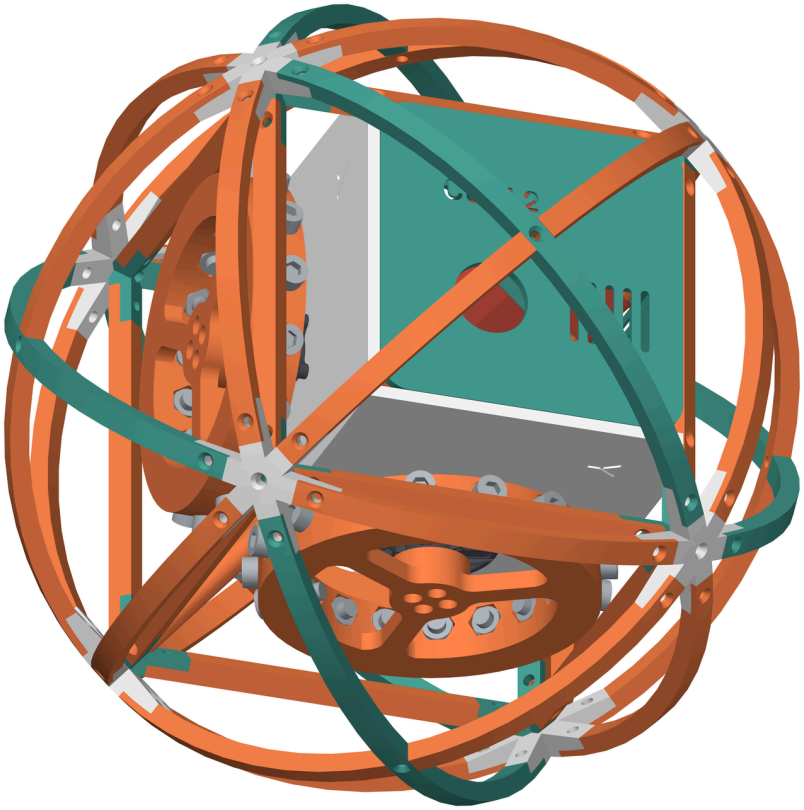
Assembly

Building the cube, step by step.



Designed by Athanasia Nikolova

All data come from tests and many real builds, not only from theory. v1.2 · November 2026

Teacher only



One thing per page

One action, one big picture. Short sentences, much white space.



Color is never the only clue

Every color also has a number, a letter or a name next to it. The part colors stay distinct with color-vision deficiency (Okabe-Ito palette).



Clear print

Instructions at 14 pt, everything else at least 12 pt, in Atkinson Hyperlegible. The text has a contrast of at least 4.5 : 1 to the page (WCAG 2.x, AA).



For a screen reader

The PDF is tagged (PDF/UA-1): headings, lists and reading order are marked, and every picture has a text alternative that is read aloud.



Print it right

In color, at 100 % - not 'fit to page'.



Pace

One page, then the action, then the next page. For a student who tires quickly: one lesson per session. The check at the end of each lesson is a good place to stop.

Safety first



Grown-up nearby

Work with your teacher or mentor in the room.



No battery yet

NO battery until the step that says so - near the end of this guide.



Fingers

The wheels are heavy. Once the cube runs, fingers away from them.



Plastic threads

Screws go into plastic: turn until it stops, then no more. Too hard and the thread is gone.



Small children

Small screws and magnets away from small children.

Before you start



Inserts guide

Every insert is in its part.



Magnets guide

The 3 magnet carriers are on the motors, the cover magnets are in.



Boards guide

The 3 boards are built and checked.



Printing guide

All printed parts are ready.

The order

- 1 Motors and wheels
- 2 Battery holder and IMU
- 3 Wiring - with the Wiring guide open next to this one
- 4 The cage
- 5 Battery and covers
- 6 The cube alone - 8 corner tips (test it)
- 7 The sphere

You need



- Hex keys 1.5, 2, 2.5, 4 and 5 mm - a ball-end 2.5 mm helps at the corners
- A 10 mm spanner (the wheel nuts)
- A tray for the screws - one cup for each kind
- A soft cloth to work on, 4 toothpicks
- The screws: sort them first (next pages)

How to read the pictures

- Color = what you fit now. Gray = already there.
- The part floats where it comes from. The arrow shows the way in.
- Under each step: what to watch, then small pictures for the tool and the way to do it.
- Each screw is drawn at its REAL size. Print this guide at 100 % and lay the screw on its picture: same length, same head - right screw.
- ×3 under the title: do the same 3 times (faces X, Y, Z).

The small pictures



by hand first



across, a little each time



level, never deeper



never force

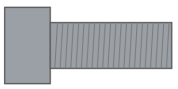


look closely

- By hand first: start every screw by hand, then the key.
- Across: 4 screws - tighten 1, 2, 3, 4, a little each time.
- Level: a countersunk head is level with the surface, never deeper.
- Never force a part. If it does not go, look why.
- Look closely before the next step.

Sort the screws

Real size. Lay each one on its picture.

54 ×		**M3 x 6** key 2 mm · DIN 7991
28 ×		**M3 x 8** key 2 mm · DIN 7991
2 ×		**M3 x 10** key 2 mm · DIN 7991
3 ×		**M3 x 12** key 2 mm · DIN 7991
3 ×		**M3 x 14** key 2 mm · DIN 7991
6 ×		**M3 x 16** key 2 mm · DIN 7991
12 ×		**M2 x 5** key 1.5 mm · DIN 912
12 ×		**M2.5 x 5** key 2 mm · DIN 912
2 ×		**M2.5 x 6** key 2 mm · DIN 912
4 ×		**M3 x 6** key 2.5 mm · DIN 912
18 ×		**M3 x 8** key 2.5 mm · DIN 912
12 ×		**M3 x 12** key 2.5 mm · DIN 912
2 ×		**M3 x 20** key 2.5 mm · DIN 912
2 ×		**M5 x 6** key 4 mm · DIN 912
48 ×		**M6 x 16** key 5 mm · DIN 912
8 ×		**M3 x 12** key 1.5 mm · DIN 914

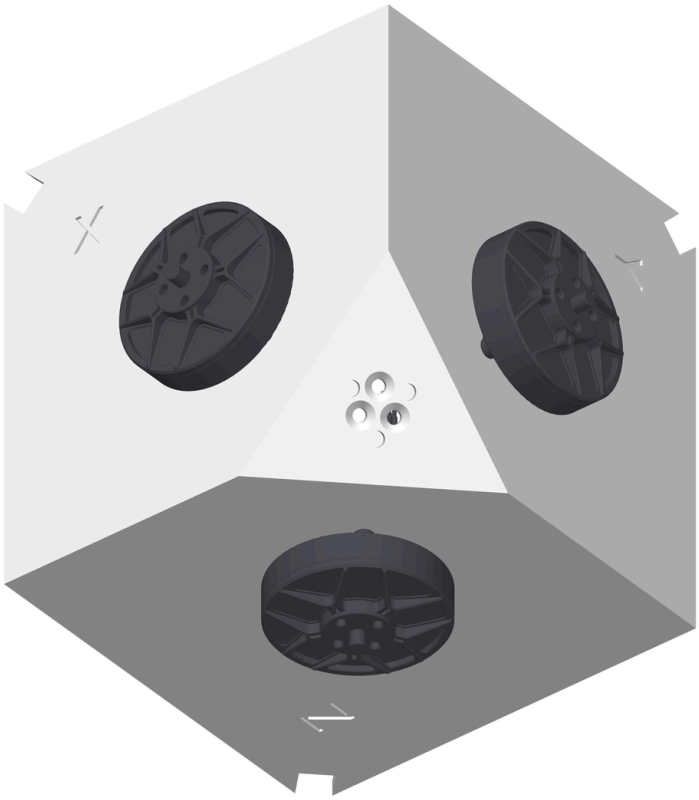
48 × 

M6 nut

spanner 10 mm · DIN 985

1 · Motors and wheels

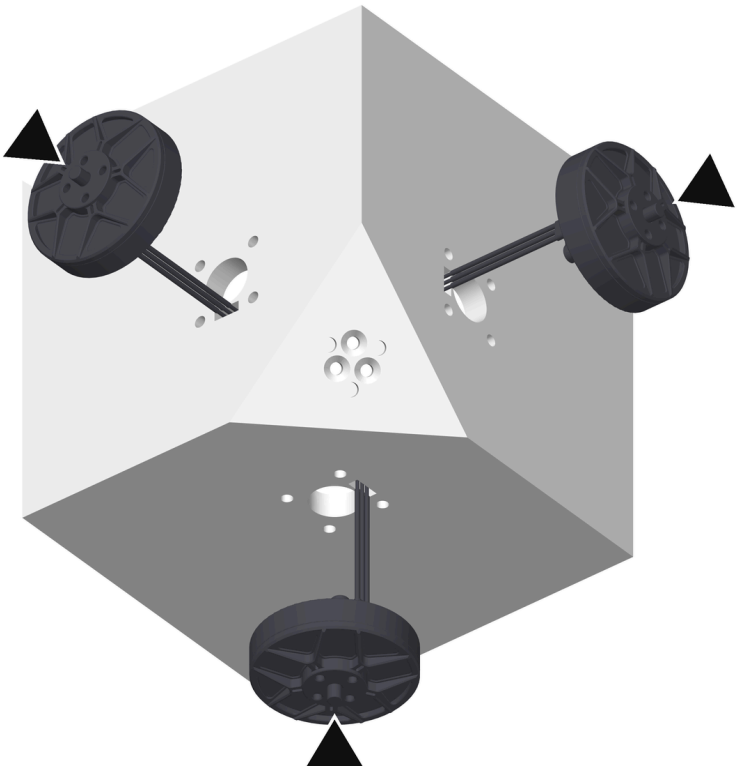
The 3 motors on the 3 closed faces, a wheel on each.



- Work on a soft cloth: the box does not slide or scratch.
- The magnet carriers are already on the motors (Magnets guide).
- Do one face completely, then the next - X, Y, Z.

Motor onto the face

×3 · faces X, Y, Z



Motor on the outside of the face, its 3 leads in through the big hole.

- Turn the motor until its 4 holes sit over the 4 holes of the wall.
- The 4 screws go in from INSIDE the box into the motor.
- Pull the leads through gently - never pinch them between motor and wall.



by hand first



crosswise



key 2.5 mm

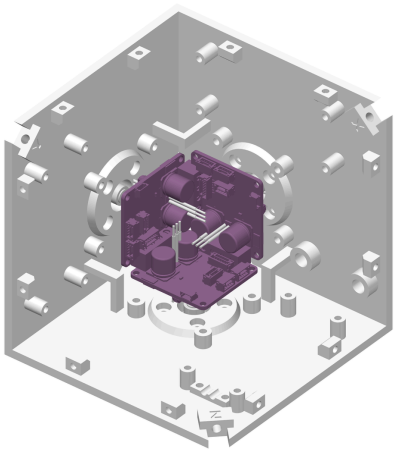
4 × 

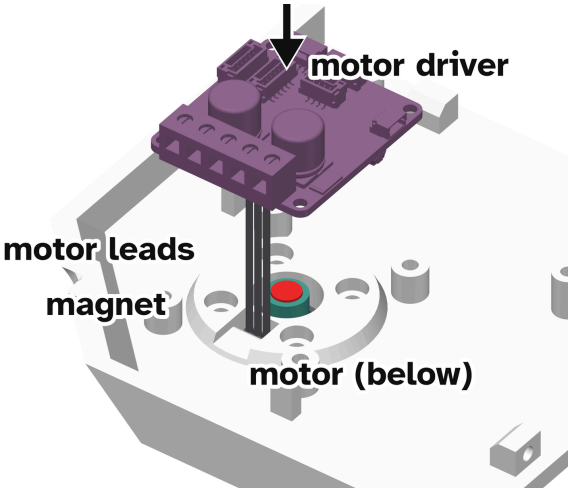
M3 x 8

key 2.5 mm

Motor driver

×3





close-up

Inside the box, over the motor's magnet: 4 screws.

- Only now: the motor is on its face and its magnet is glued on (Magnets guide). Its 3 leads stand up beside the board.
- The small sensor in the middle of the board sits right over the magnet.
- Look from the side: a small gap between magnet and board. Nothing touches.
- The 3 motor leads into its U, V, W terminals - in any order: the driver finds the order itself when it calibrates (software).


the gap


no force


key 2 mm

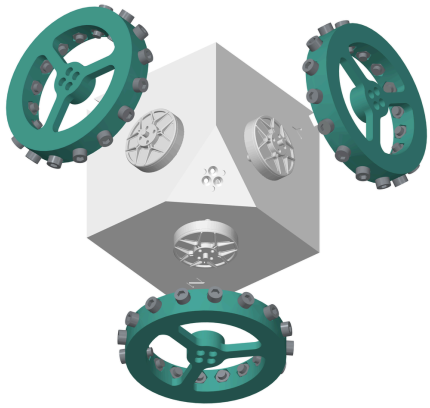
4 × 

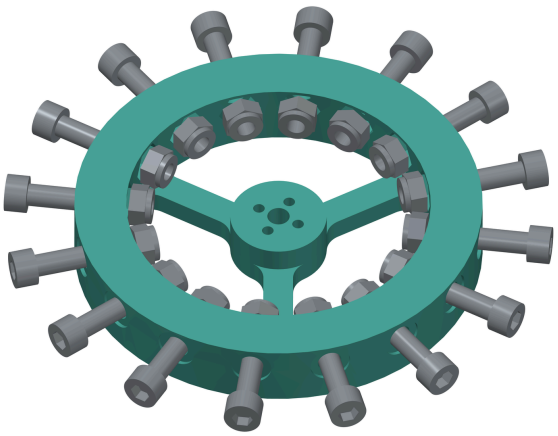
M2.5 x 5

key 2 mm

Rim masses into the wheel

×3 · 3 wheels





close-up

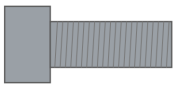
16 screws through the rim, a nut on each, from the inside.

- All 16 the same way: head outside, nut inside - the wheel stays balanced.
- The nyloc nut turns hard - that is normal. Hold the head with the key, turn the nut with the spanner.
- Tight: the masses spin fast, they must not move.


key 5 mm


spanner 10


on the table

16 × 

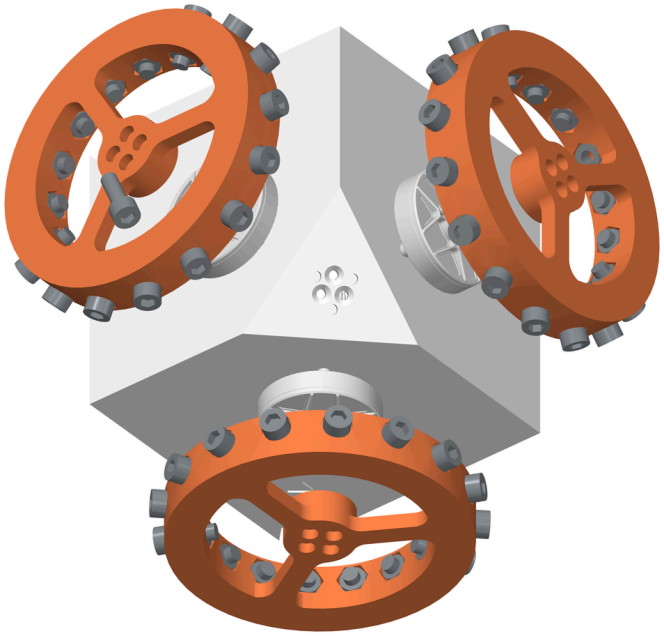
M6 x 16
key 5 mm

16 × 

M6 nut
spanner 10 mm

Wheel onto the motor

×3



The thin metal ring on the motor, then the wheel, 4 screws.

- The ring sits centered on the turning part of the motor.
- The wheel's 4 holes over the 4 holes under the ring.
- Turn the wheel by hand: round and free, it never touches the box.



by hand first



crosswise



turn and look

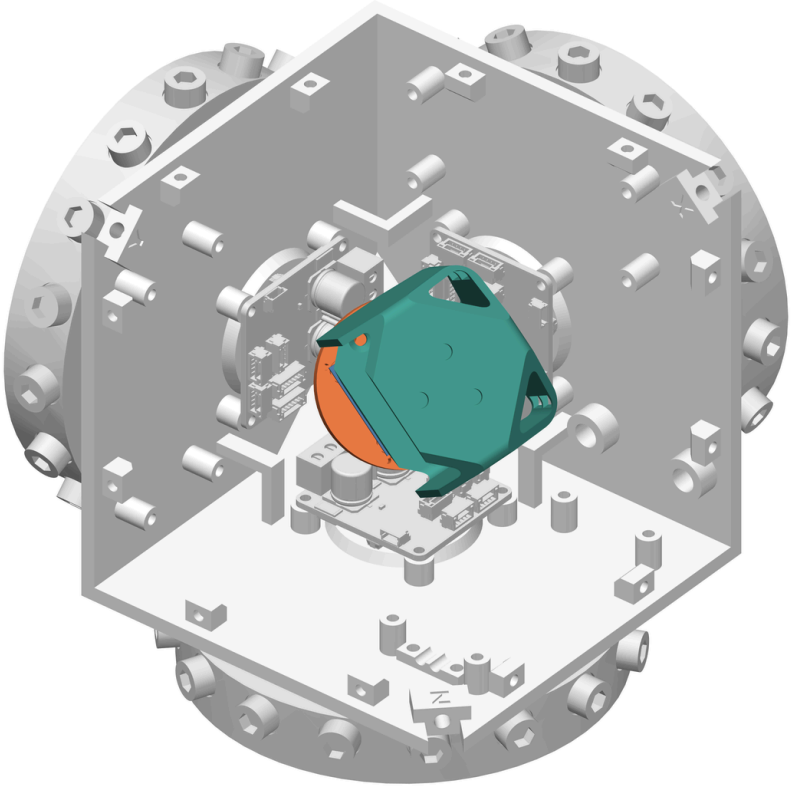
4 × 

M3 x 12

key 2.5 mm

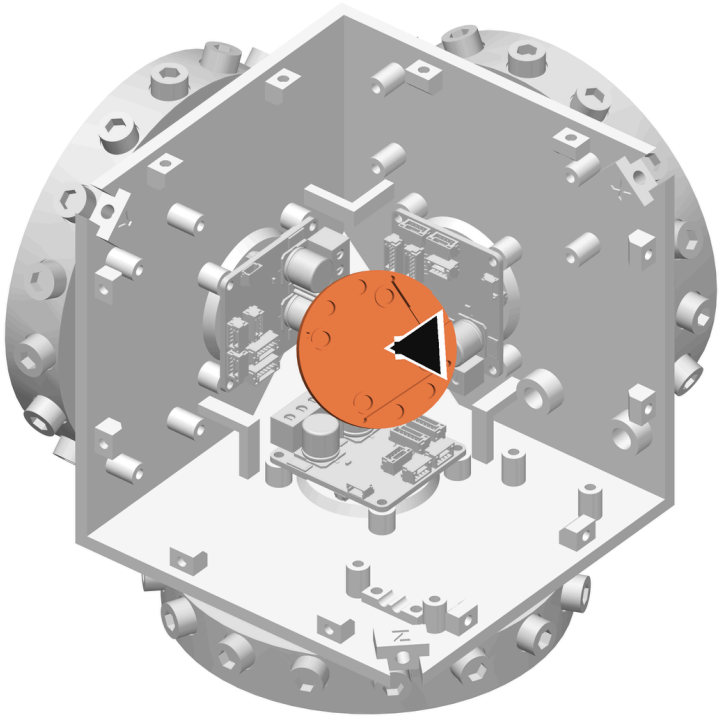
2 · Battery holder and IMU

Along the long diagonal, from corner 0 - the corner where the 3 motor faces meet.



- The IMU tells the cube how it is tilted. Its position matters: read its step twice.
- The straps go through the basin BEFORE the basin goes in.

Battery tube to corner 0



Tube in from inside, its foot on corner 0. 3 screws from OUTSIDE.

- Turn the tube until its 3 holes sit over the 3 holes in the corner.
- Countersunk heads: level with the outside, not deeper.


by hand first


flush

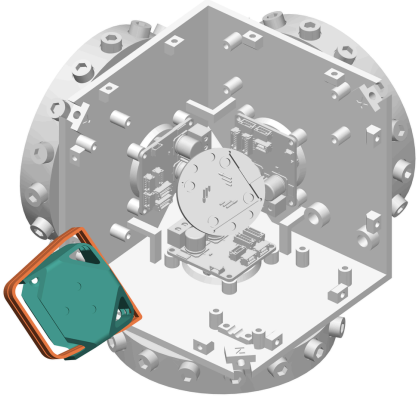

key 2 mm

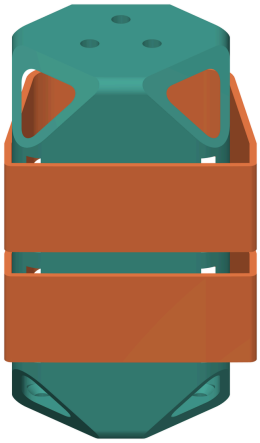
3 × 

M3 x 12

key 2 mm

Straps through the basin





close-up

The 2 hook-and-loop straps through the slots in the back of the basin.

- Now, while the basin is in your hands.
- Each strap through its own pair of slots.
- Leave them open: they close round the battery at the end.

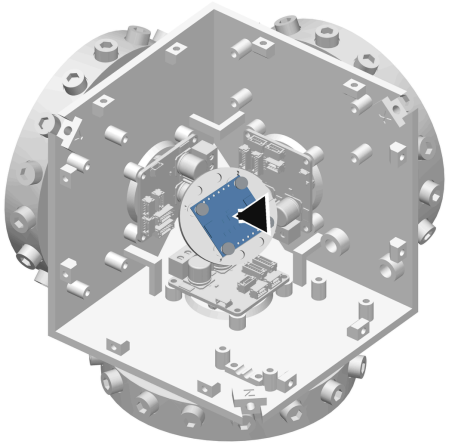


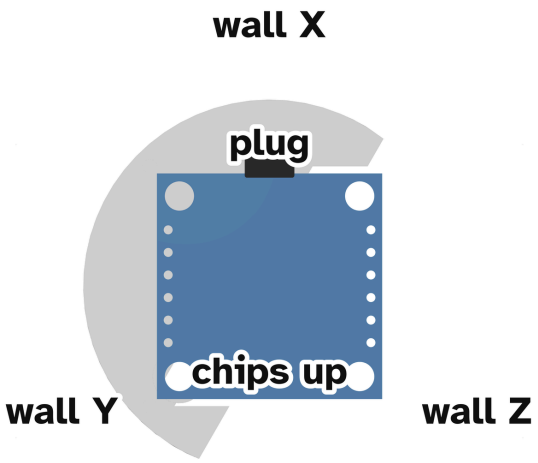
2 straps



on the table

IMU into the table





close-up

The IMU into the pocket of the table, 4 screws M3x6.

- The IMU's parts (its chips) face UP - toward the basin and the middle of the cube. Its 4-pin plug points to the wall with the letter X, into the table's slot.
- The board sits on the pocket's 4 small bosses. The pocket's corners are rounded out: it goes in by hand.
- Tighten across, a little each time: the board must not bend.


plug to X


no force



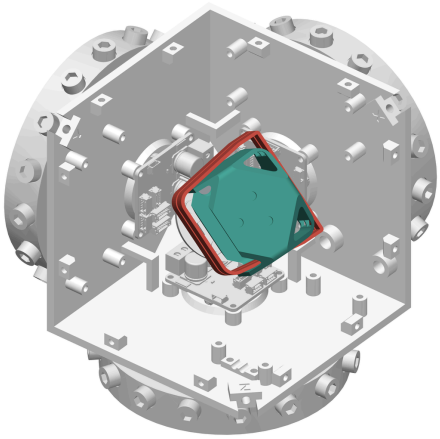
crosswise

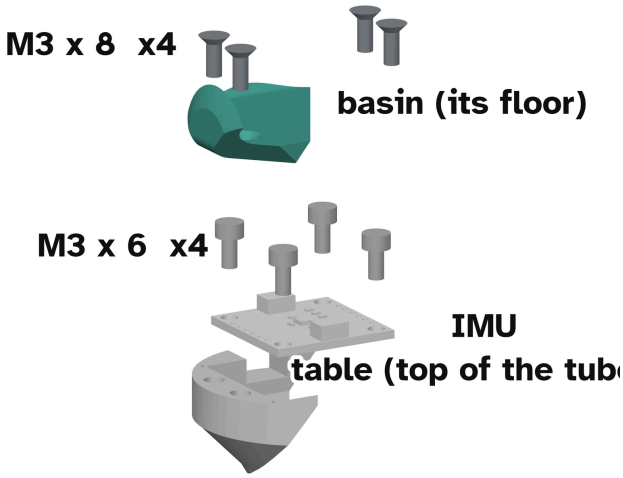
4 × 

M3 x 6

key 2.5 mm

Basin onto the table





close-up

The basin on the table, 4 countersunk screws M3x8 from inside the basin.

- The basin's floor lies flat on the table: nothing touches the IMU.
- The straps are already through the basin (Straps through the basin).
- Heads level with the floor - the battery lies on it. Tighten across.



crosswise


flush


key 2 mm

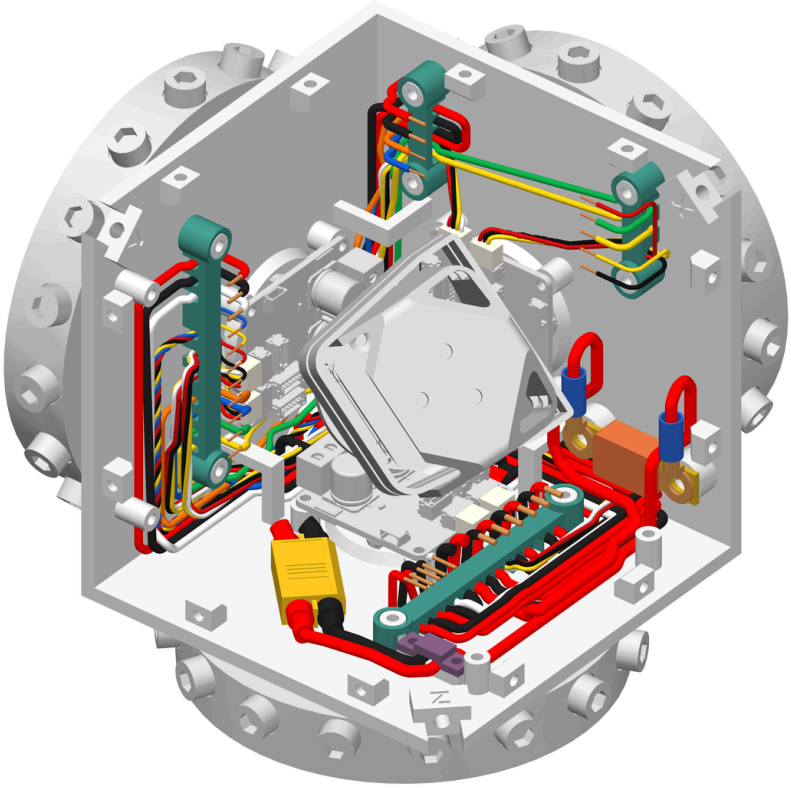
4 × 

M3 x 8

key 2 mm

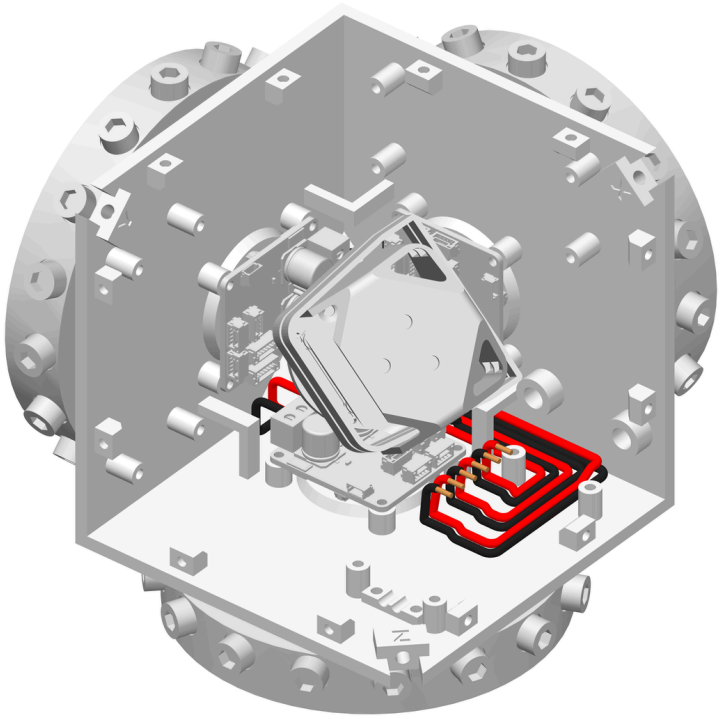
3 · Wiring

All the wires, in 5 stages. Each wire, step by step, is in the Wiring guide: work with it open next to this one.



- The Wiring guide has its own safety page, cut lists and checks.
- Still NO battery.
- Lowest wires first: each wire lies on the ones under it.

Motor power



6 thick wires from the 3 motor drivers to board Z.

- They lie flat on the floor, side by side, red and black in turn.
- Each wire where its arch of the cable clamp will be. The clamps and the boards go on later, over the wires.
- Wiring guide: chapter 1.

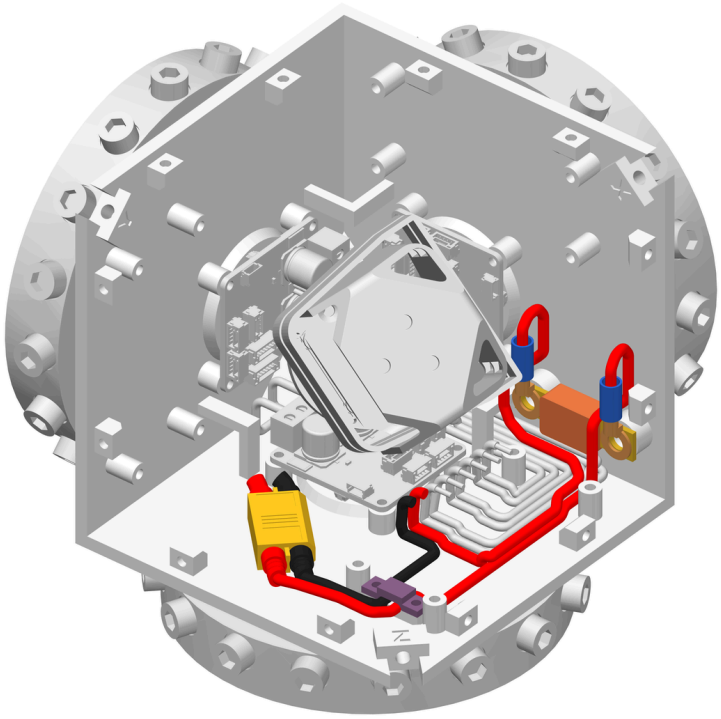


strip 7 mm



one per arch

Battery supply



Fuse, battery connector and cable clamp.

- The fuse's 2 lugs on their M5 screws, the clamp over the thick wires.
- The battery's own half of the connector: teacher only.
- Wiring guide: chapter 2.



fuse solder heat-shrink

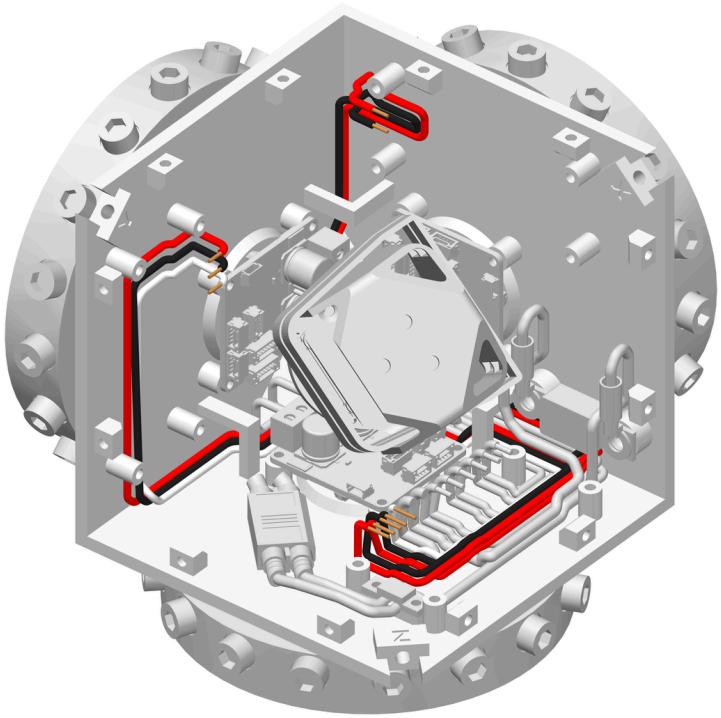
2 ×  **M5 x 6**

key 4 mm

2 ×  **M2.5 x 6**

key 2 mm

Board power



The 5 V wires from board Z to boards Y and X, and SENSE.

- They lie on top of the motor wires.
- Wiring guide: chapter 3.

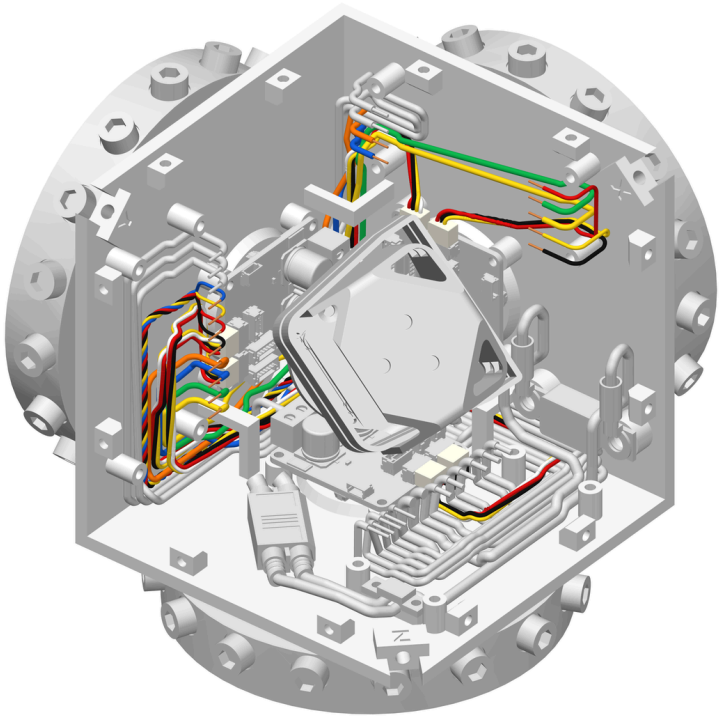


strip 7 mm



one per arch

Signals



The links, the CAN cables and the I2C cable.

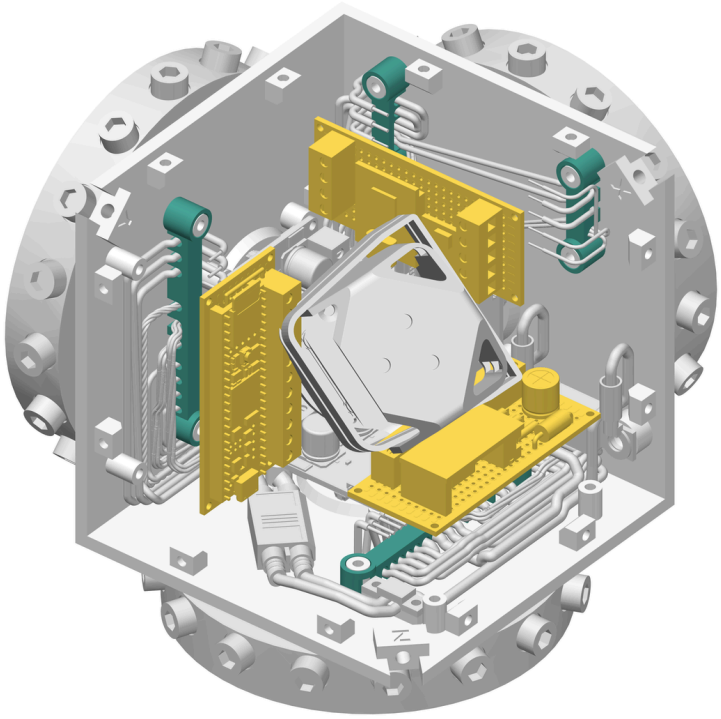
- Thin wires: never pull them tight round a corner.
- Wiring guide: chapter 4.


signals


no pulling

Boards on

×3 · 4 each



The 4 clamps over the wires, a spacer on each ring, the 3 boards on their standoffs.

- 4 screws per board. Board Z first, then Y, then X.
- Each wire end now stands in front of its terminal. Do not connect yet.

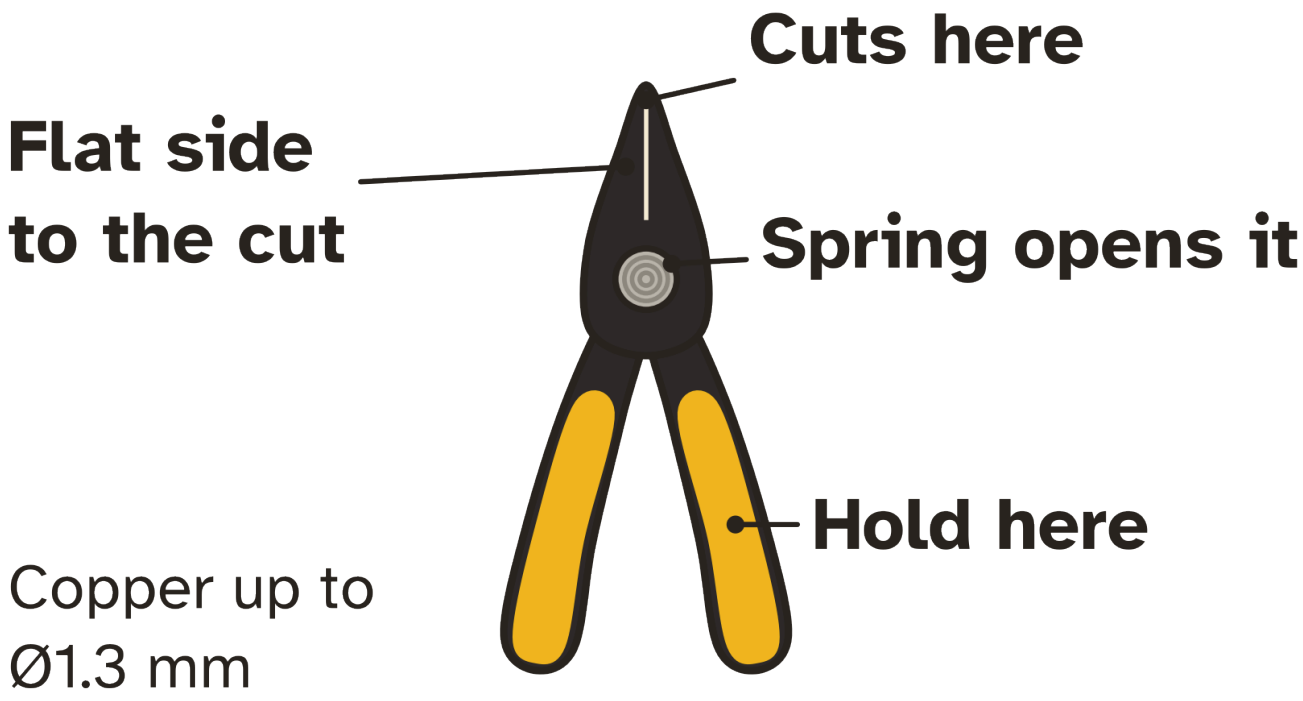
key 2 mm

by hand first

12 × **M2 x 5**

key 1.5 mm

Mark and cut



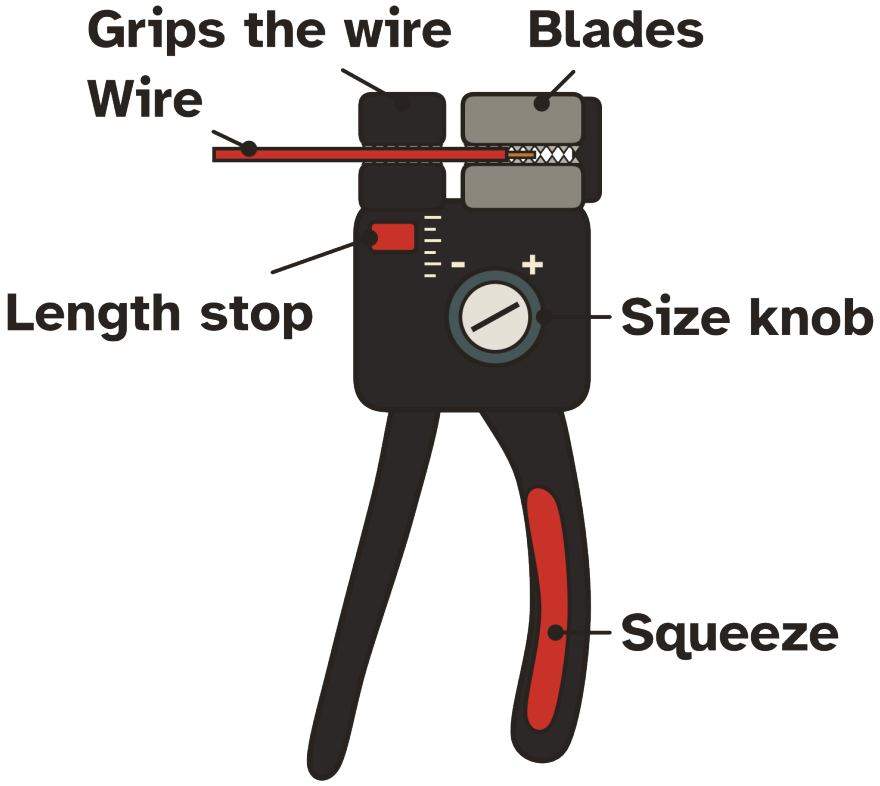
Each wire end: a mark with a permanent marker 7 mm past its terminal's opening, cut there.

- Hold the wire against its terminal to find the place. One wire at a time.
- The side cutter: its flat side toward the cut.
- The pole name (z1, y3 …) is on the wire's name flag and on the terminal map.


marker


side cutter

Strip 7 mm



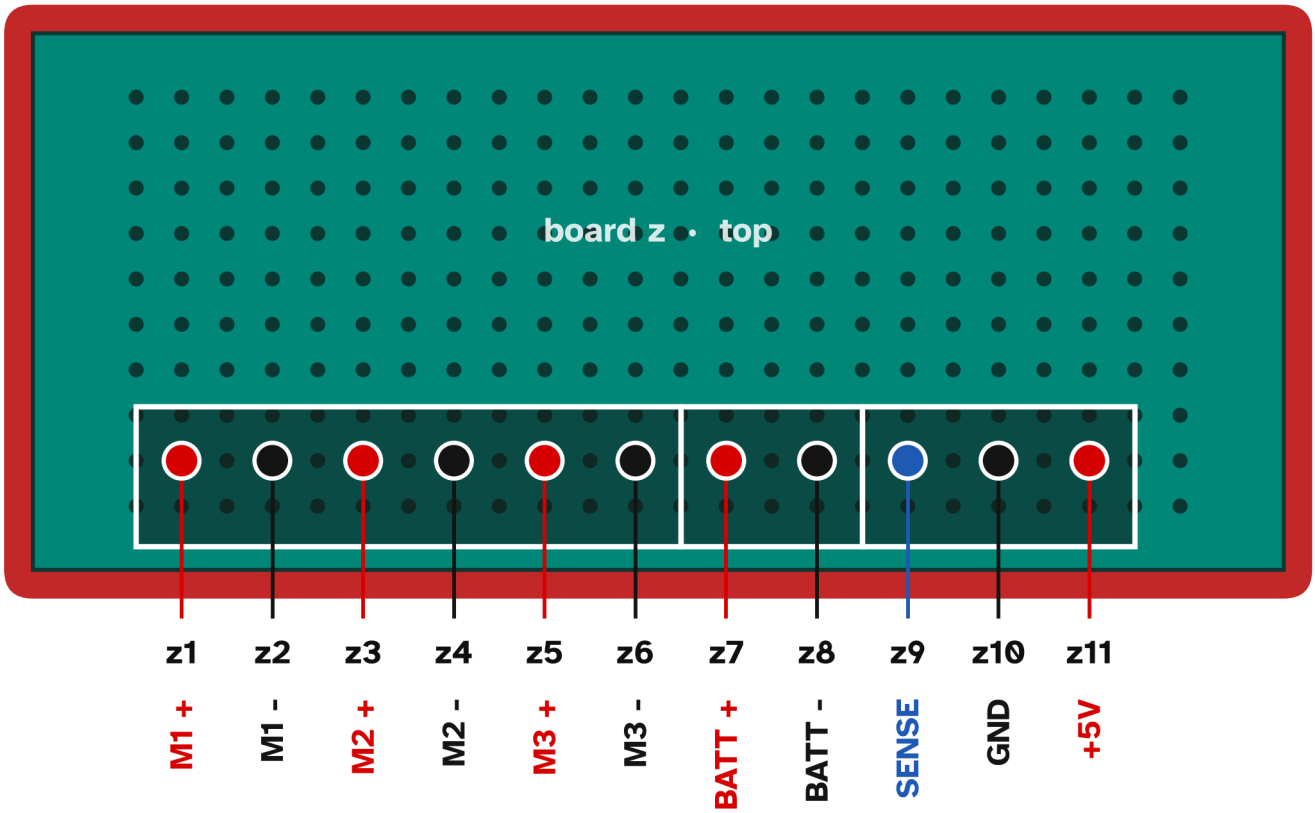
The wire stripper set to the wire size, the wire in up to 7 mm, squeeze.

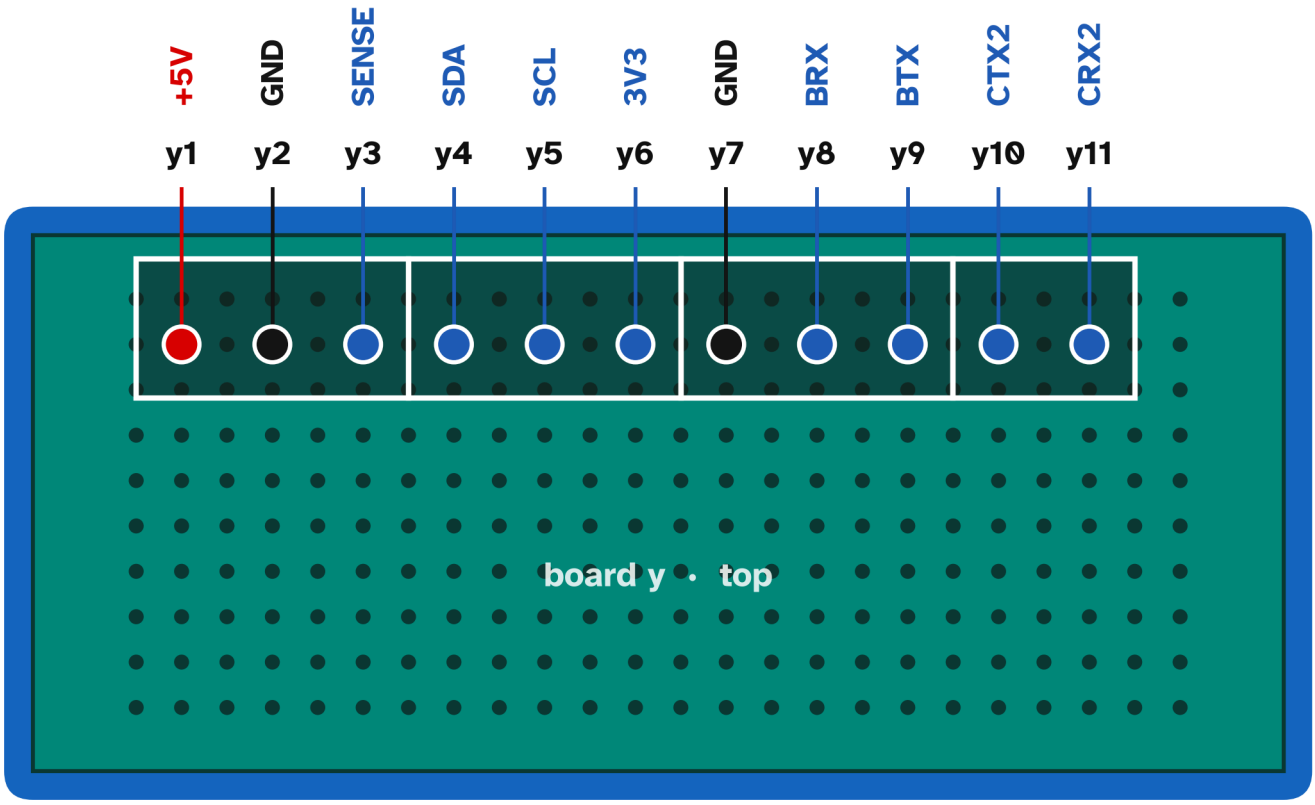
- 7 mm of bare copper - no more.
- Stranded wire: twist the strands tight, no solder on them. Solid wire: straight as it is.


7 mm


no nicks

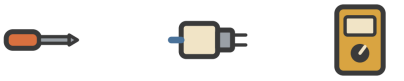
Into its terminal



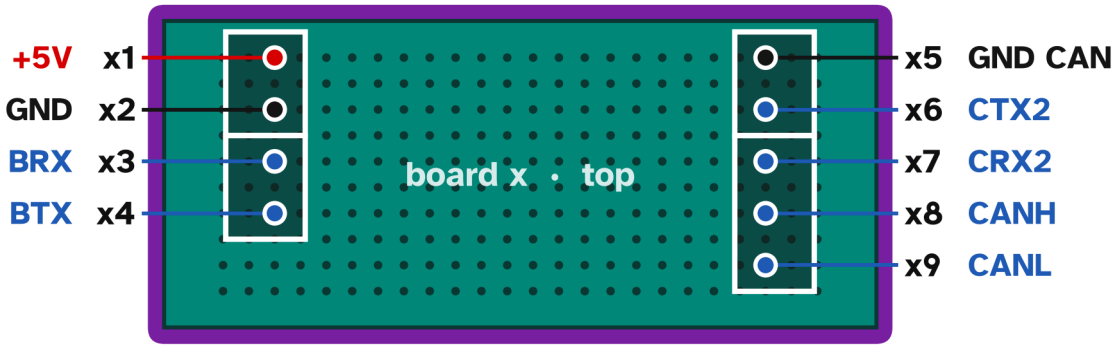


Each end into the pole of its name - the maps: board Z here, Y beside it, X below.

- All the copper in, none outside. Tighten with the small screwdriver.
- Pull gently: it must not move. Then the plugs: CAN into the drivers, I2C into the IMU.
- Final check with the multimeter - in the Wiring guide. Still NO battery.



Board X terminals



Board X: the two blocks, one on each side.

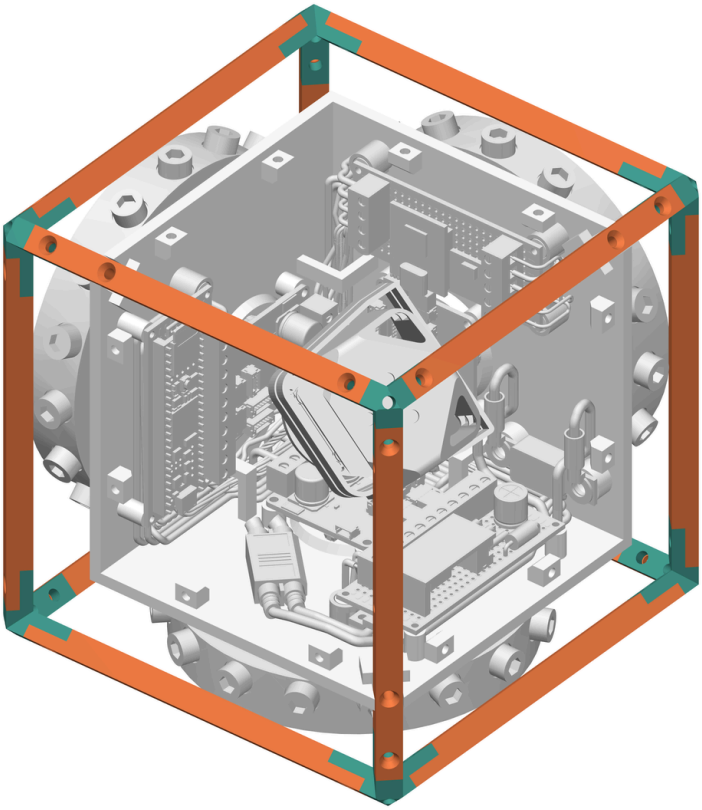
- x1 - x4 on the west block, x5 - x9 on the east block.



terminals

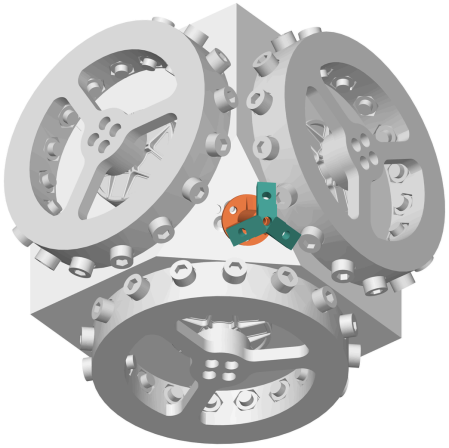
4 · The cage

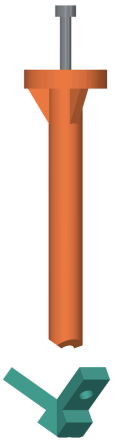
8 corners, 12 beams. It holds the box at 2 corners and carries the sphere.



- Corner 0 and the far corner (1,1,1) are the 2 corners with 2 inserts.
- Countersunk screws: level with the surface, never deeper.
- Every cage screw snug first; tighten them all when the frame is closed.

Box standoff into corner 0





close-up

On the table: the long screw down through the standoff into the corner.

- Take a corner with an insert INSIDE (one of the 2 corners with 2 inserts).
- The head goes right down into its round pocket - level with the end of the standoff.
- Tight: the standoff must not turn.



on the table



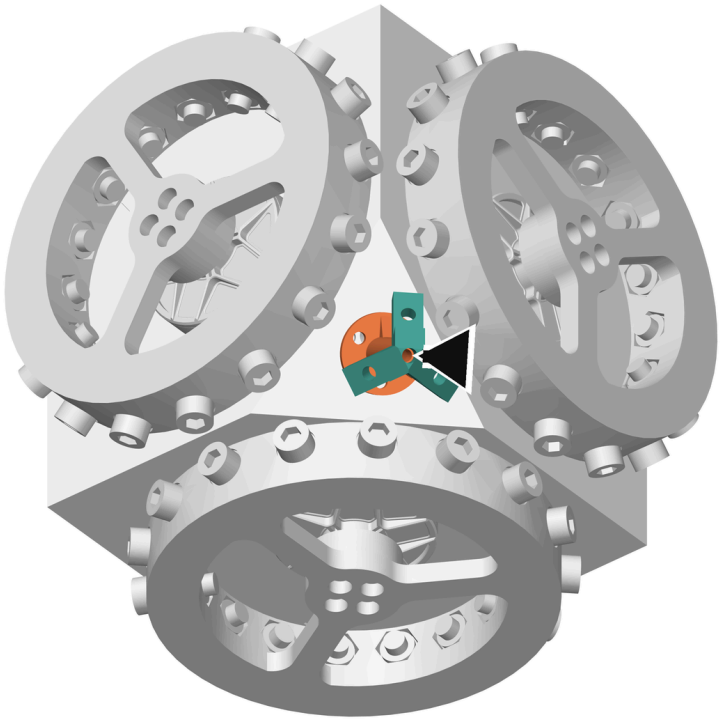
long key 2.5

1 × 

M3 x 20

key 2.5 mm

Corner 0 onto the box



The round flange flat on corner 0 of the box. 3 screws through the flange.

- Nothing may lie under the flange.
- Its 3 screws sit between the 3 heads of the tube screws.
- A ball-end key reaches past the cage corner.


by hand first



across

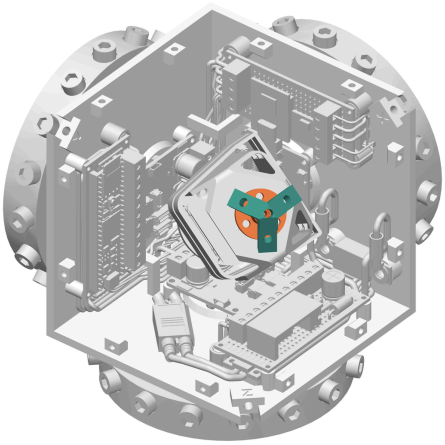

ball-end 2.5

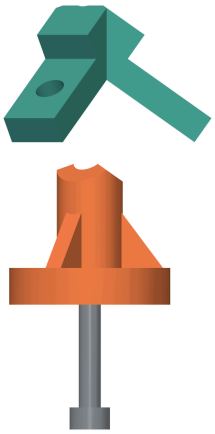
3 × 

M3 x 8

key 2.5 mm

Hub standoff into the far corner





close-up

On the table: the long screw up through the hub standoff into the 2nd corner with 2 inserts.

- The top of the standoff has the shape of the corner: turn it until it sits.
- Head level with the flange.


on the table


it sits

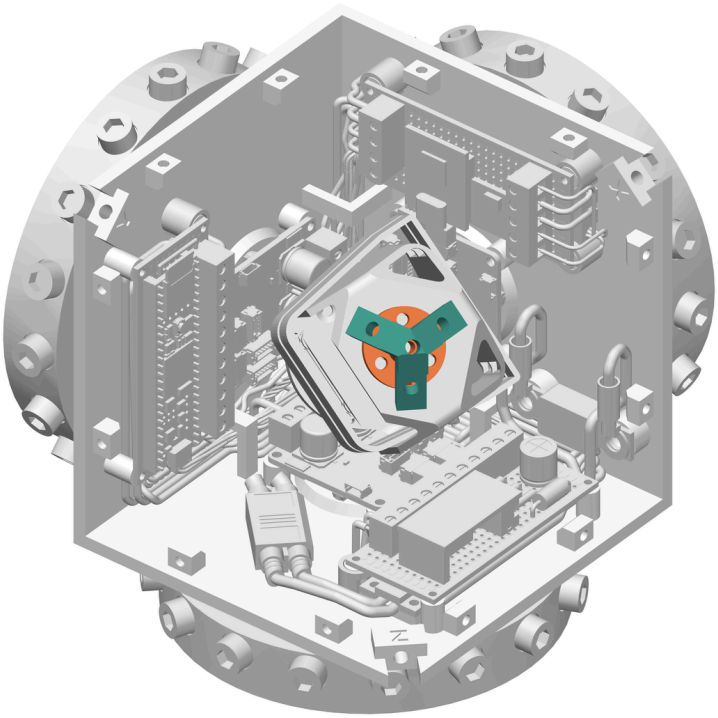

long key 2.5

1 × 

M3 x 20

key 2.5 mm

Far corner onto the basin



Through an open face: the flange flat on top of the basin, 3 screws.

- No covers yet - work through the open faces.
- Do not pull on the basin: the IMU stack holds it.


by hand first


no pulling

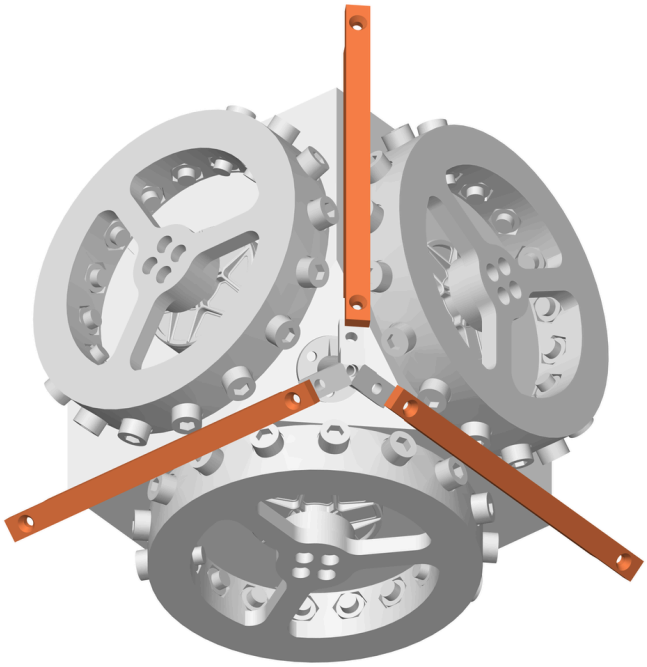

key 2.5 mm

3 × 

M3 x 8

key 2.5 mm

3 beams at corner 0



Each beam lies on the corner from the outside. 1 screw each.

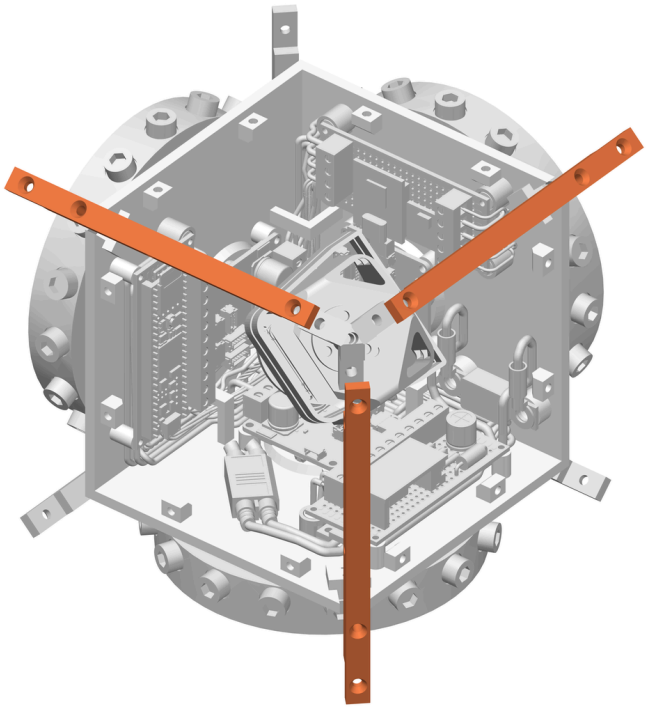
- Plain beams here (no magnets).
- Snug only - tighten when the frame is closed.



3 × **M3 x 8**

key 2 mm

3 tie beams at the far corner



The 3 beams with magnets: 1 screw into the corner, the long one into the box.

- Their magnets face the covers.
- The long screw (M3 x 14) goes through the beam into the box wall.

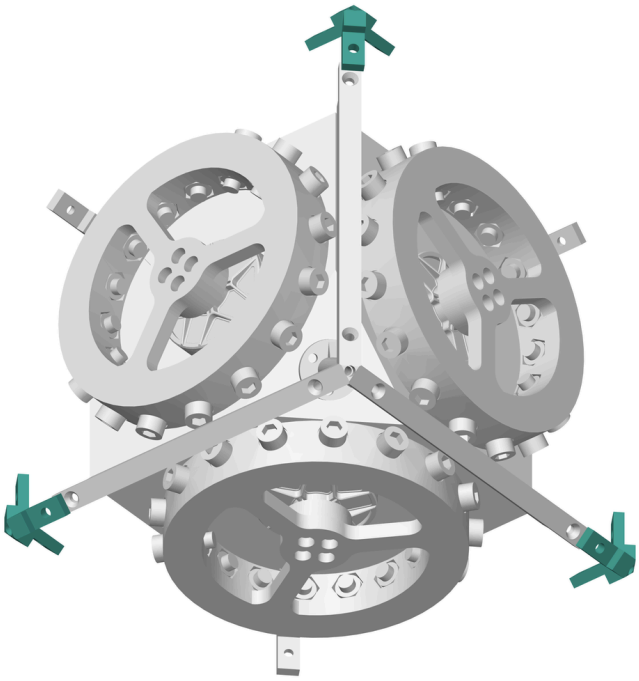

magnets in


flush

3 × 
M3 x 8
key 2 mm

3 × 
M3 x 14
key 2 mm

3 corners on the corner-0 beams



Each corner closes on the free end of a beam. 1 screw each.

- Snug only.



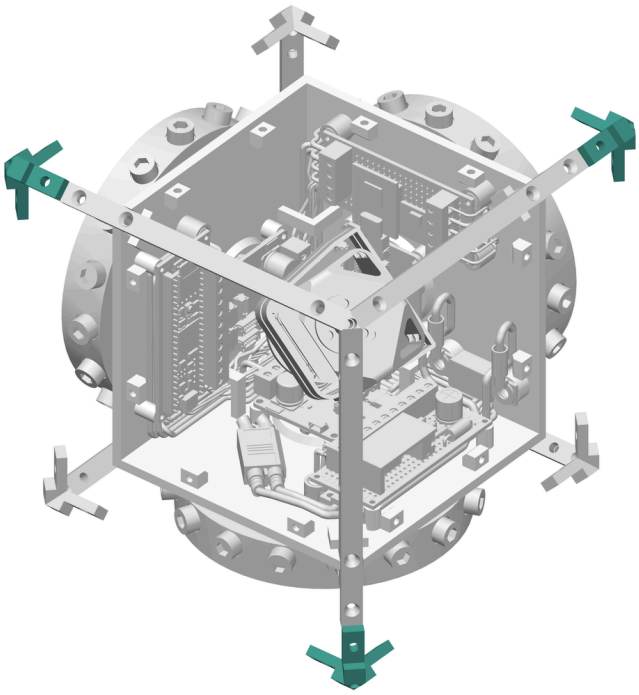
snug

3 × 

M3 x 8

key 2 mm

3 corners on the tie beams



Each corner closes on the free end of a tie beam. 1 screw each.

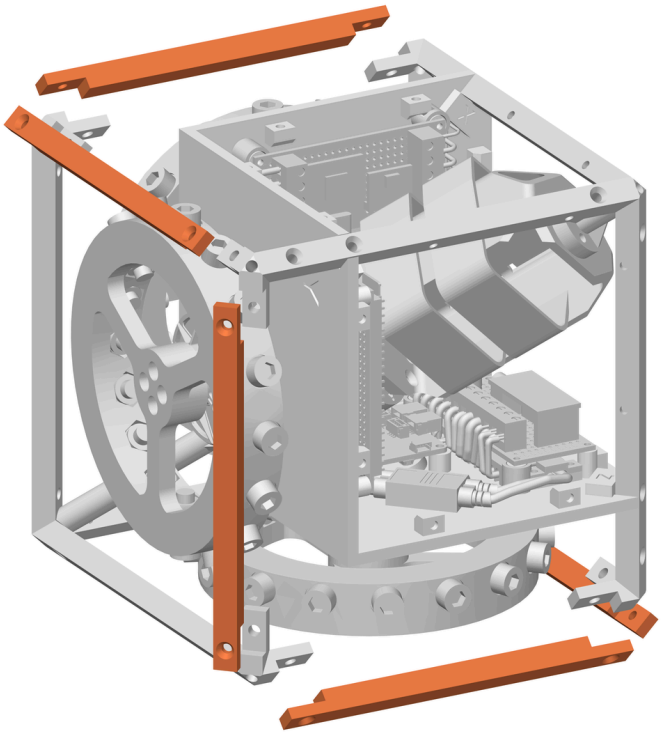
- Snug only.

snug

3 × **M3 x 8**

key 2 mm

The last 6 beams



Each between two corners, 2 screws.

- Now tighten every cage screw.
- Shake gently: nothing moves.

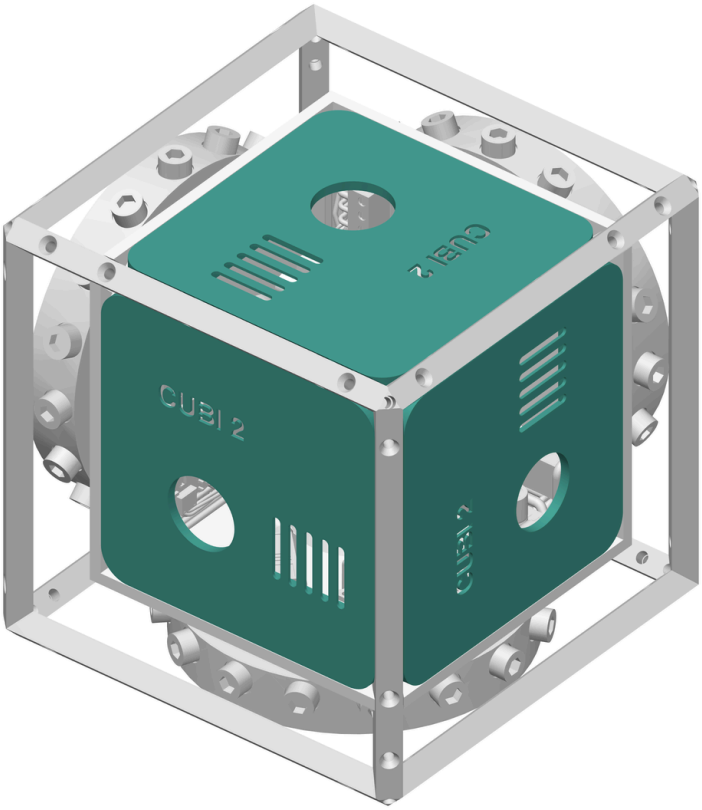
flush check

12 × **M3 x 8**

key 2 mm

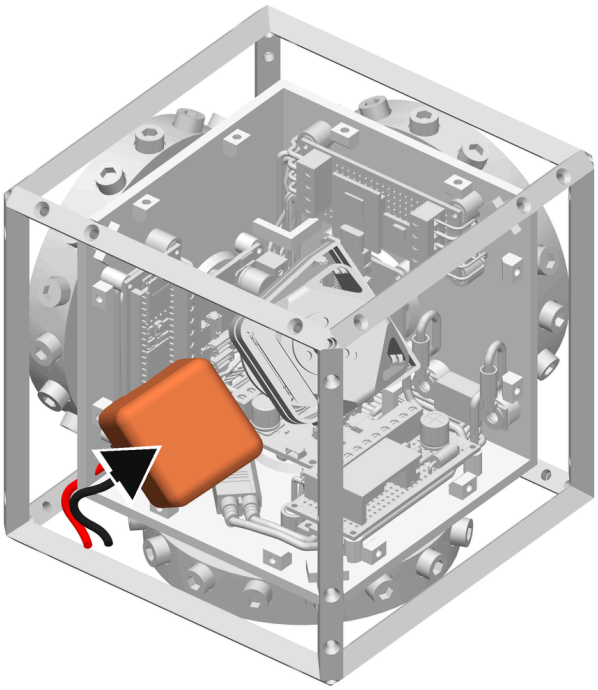
5 · Battery and covers

Only with your teacher.



- Never force the battery. Never let its two leads touch.
- A battery that is puffy, hot or damaged is not used.

Battery in



Slide the battery in from the X side, across the diagonal, under the open straps.

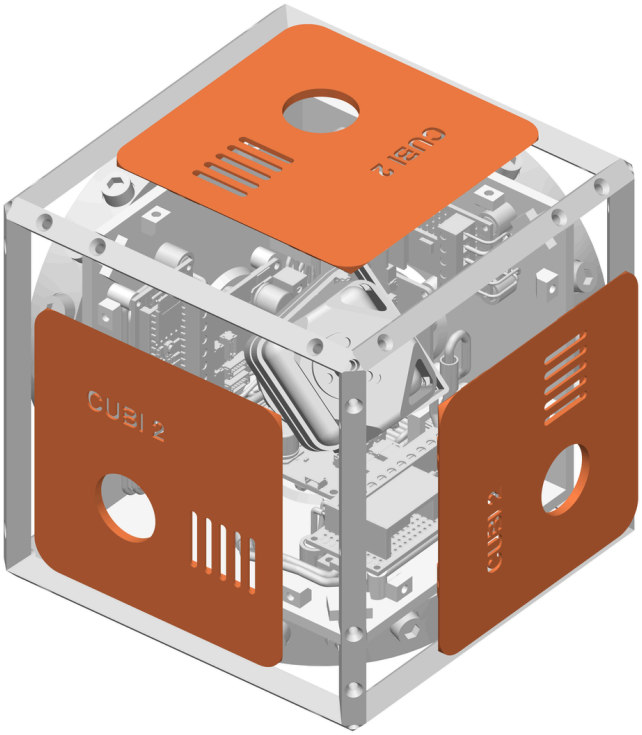
- Its connector end first, toward corner 0.
- Pull each strap tight round the battery and close it: the straps hold it in the basin.
- Plug it in only together with your teacher: the first power-up (Wiring guide, final check). The connector fits only one way: red + to +. Never force it.


straps tight


no force


with teacher

Covers on



Each cover on its own face. The magnets pull it in.

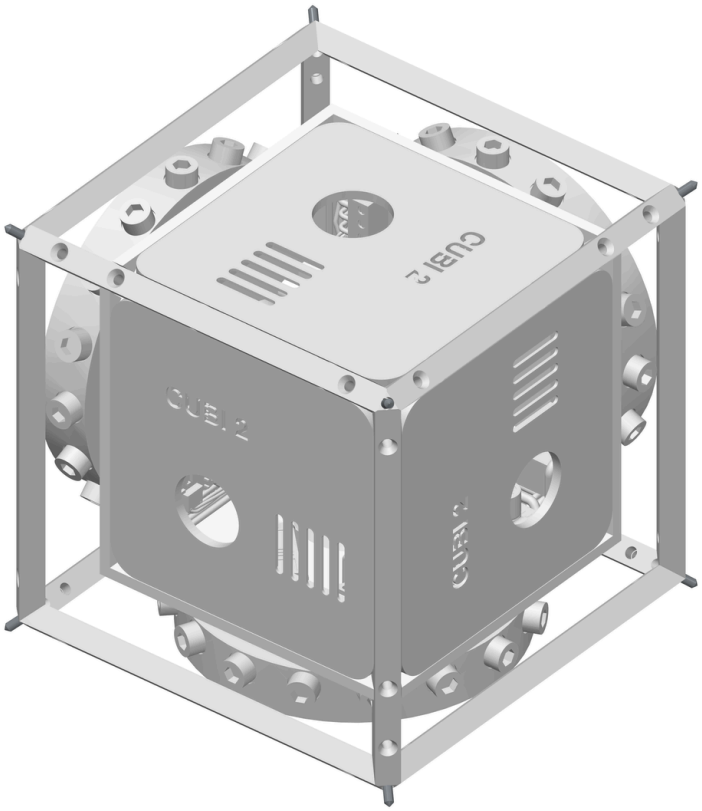
- Its letter (X, Y, Z) inside matches the box.
- Look first: no wire on the edge - the cover would pinch it.
- To take a cover off: pull at its edge.


no wire


no force

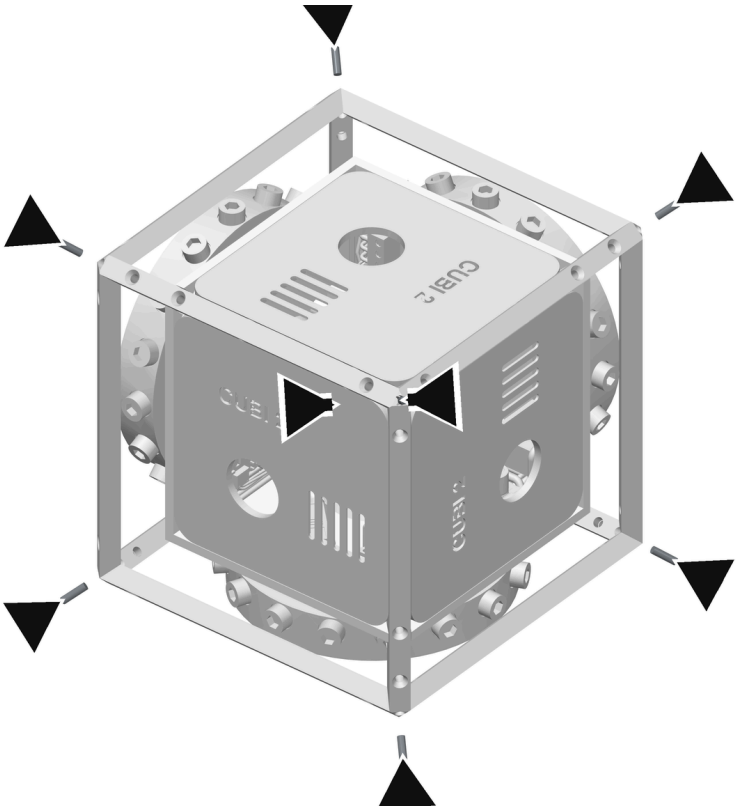
6 · The cube alone

A stage to test it: the bare cube balances on a corner. 8 corner tips, no sphere yet.



- The tips go into the same inserts as the navels later.
- Test it on a soft mat: the cube can fall over.

Tips on the 8 corners



One grub screw with a cone point into the tip insert of each corner.

- The point sticks out 6.2 mm from the cut face - measure with the depth gauge of the caliper. Then the point is exactly where the corner would be: the cube still sits flat on a face.
- Less: it does not reach the ground. More: the cube rocks on 4 tips.
- Before the sphere (chapter 7): all 8 tips come out again.


key 1.5 mm

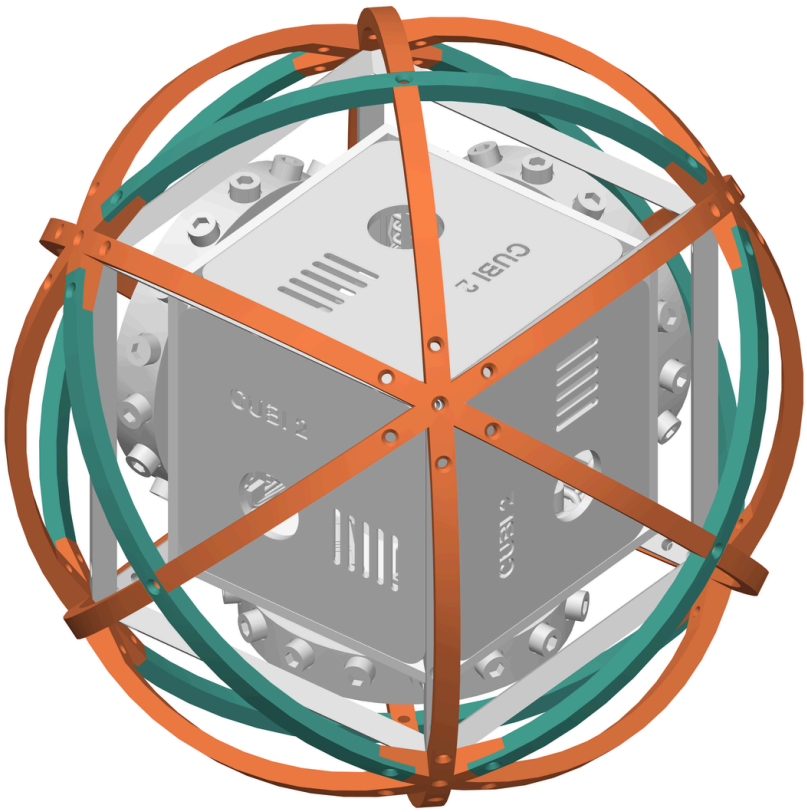

6.2 mm out


soft mat

8 × 
M3 x 12
key 1.5 mm

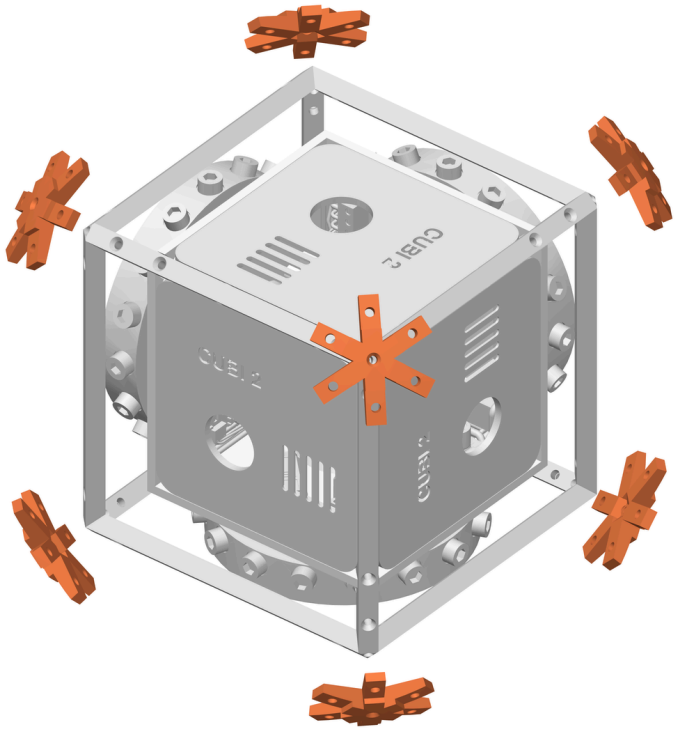
7 · The sphere

8 navels and 24 arcs make the ball the cube rolls in.



- Lay the cube on a soft cloth: the arcs scratch.
- The order: navels, edge arcs, lower diagonal arcs, upper diagonal arcs.
- All screws snug first; tighten them all when the ball is closed - it comes out round.

Navels on the 8 corners



One screw in the middle of each navel, into the corner tip.

- First the 8 corner tips out (chapter 6): the navels go into the same inserts.
- Corner 0 and the far corner take the SHORT screw (M3 x 10): their inside is taken.
- Turn each navel so its arms point along the edges and the face diagonals.

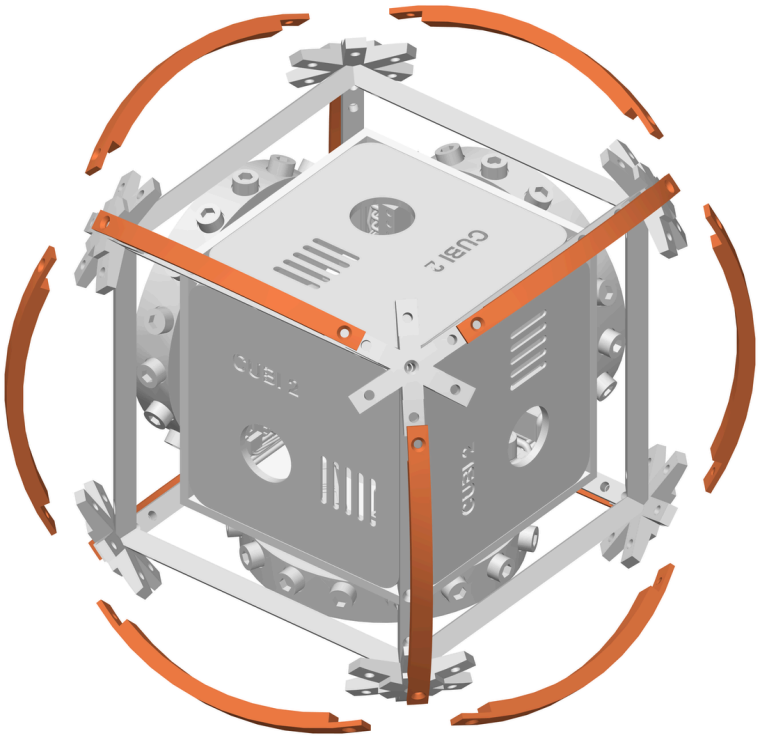

soft cloth


flush

6 × 
M3 x 16
key 2 mm

2 × 
M3 x 10
key 2 mm

Edge arcs



12 arcs, one over each edge. 1 screw at each end.

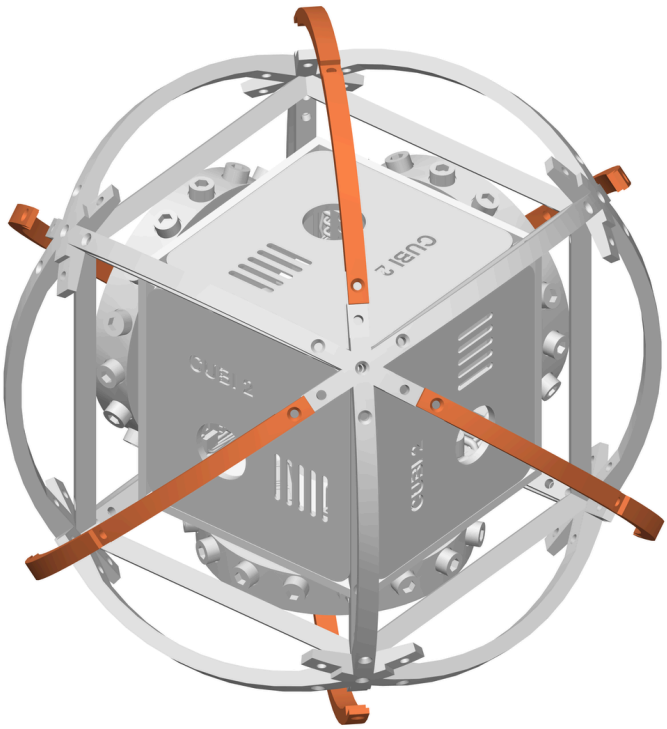
- Each end into a navel arm.
- Snug only.

snug flush

24 × **M3 x 6**

key 2 mm

Diagonal arcs - the lower ones



On each face, one diagonal arc first: the LOWER one.

- The lower arc has an insert in the middle.
- 1 screw at each end, snug.


insert up


snug

12 × 

M3 x 6

key 2 mm

Diagonal arcs - the upper ones



The second arc crosses over the first. 1 screw at each end, 1 in the middle.

- The middle screw goes into the lower arc's insert.
- Now tighten every sphere screw. Roll the ball: it is round.



all tight


round?

12 ×  **M3 x 6**

key 2 mm

6 ×  **M3 x 6**

key 2 mm

Final check



Every screw

Tight, no more.



Each wheel

Turn it by hand: free, no rubbing, no click.



Shake it gently

Nothing rattles inside.



Roll it on the floor

The sphere is round, no arc sticks out.



Spare screws left over?

Find where they belong before the first run.